**TMJC J2H2 Geography Prelim**
**Suggested Response**

Answer **both** questions

**Cluster 1: Development, Economy and Environment**

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| **1** | | Resource 1 shows the GDP growth of selected ASEAN countries.<br>Resource 2 shows US tariffs on different ASEAN countries, with proposed additional tariffs.<br>Resource 3 shows ASEAN nations exports to major destinations<br>Resource 4 shows 2023 SDG levels and trends for ASEAN countries on select SDG indicators. | |
| | **(a)** | Using Resource 1, describe the changes in GDP growth for all the selected ASEAN countries from 2024 to 2025. | [4] |
| | | • Half of the countries decreased while half increased,<br>• Vietnam, Malaysia, Singapore decreased<br>• Philippines, Indonesia, Thailand increased<br>• Vietnam largest decrease by 0.5%<br>• Philippines largest increase 0.6% | |
| | **(b)** | Suggest reasons for the differing changes in GDP growth for the ASEAN countries in Resource 1. | [4] |
| | | *any two possible arguments accounting for countries increase and decrease in GDP growth<br>• Vietnam, Malaysia, Singapore – labour too costly (change in comparative advantage) , loss of market share for selected products in the market<br>• More preference for Philippines and Thailand products and services/tourism | |
| | **(c)** | Using Resources 2 and 3, explain how the proposed additional tariffs can have differing impacts on ASEAN countries. | [6] |
| | | Differing impacts +ve and -ve economic impact<br>• Vietnam -ve economic impact, reduce exports to USA<br>• Vietnam – factories might close down as well - -ve social impacts<br>• Singapore +ve economic impact, might have more exports<br>• Countries with lower tariffs may experience +ve social impact | |

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| | **(d)** | Explain what states can do to mitigate the impact of proposed additional tariffs by US as shown in Resource 2. | [6] |
| | | • Sign more FTA / bilateral trade agreement with other countries – find new trading partners<br>• Set up special economic zones / export processing zones to attract other TNCs<br>• Negotiate through regional blocks like ASEAN and/or encourage more intra-regional trade | |
| | **(e)** | Using Resource 4, compare SDG levels and trends for Singapore and Malaysia. | [4] |
| | | • In terms of levels, Singapore doing worst as Singapore have 3/6 with major challenges but Malaysia only 1/6<br>• In terms of SDG 9, Singapore is doing much better than Malaysia as it has SDG achieved while Malaysia is major challenges<br>• For trends Singapore is much better with 3/6 on track or maintaining SDG achievement<br>• For SDG 11, Singapore and Malaysia have same level and trends | |
| | **(f)** | Account for the difference in the SDG trends between Singapore and Malaysia | [6] |
| | | • Structure of the economy and GDP per capita can reflect SDG 9<br>• Singapore has clean water but no water resources so it is a major challenges compared to Malaysia with ample water resources<br><br>• Responsible consumption and production is much worse level in Singapore due to affluent society and high generation of waste<br>• Climate action in Singapore facing major challenges also due to high energy usage | |

## Cluster 2: Tropical Environments

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| **2** | Resource 5 shows the map and climograph of Dhaka, the capital city of Bangladesh.<br>Resource 6 temporal patterns in river discharge measured at a research station situated at the lower course of the Padma River in Bangladesh.<br>Resource 7 shows the forecast of Tropical Cyclone Remal in May 2024.<br>Resource 8 shows land cover maps of Dhaka metropolitan area from 1989 to 2009. | | |
| | **(a)** | Describe the climatic characteristics of Dhaka, Bangladesh as seen in Resource 5. | [4] |
| | | High average annual at 25.3 °C<br>Distinct seasonality<br>- temp range of about 9°C; warmest in April, coolest in end -Jan<br>- warm months from March to October; cooler months from November to Feb<br><br>High rainfall totals: 2055mm<br>Distinct seasonality<br>- highest in June at 380 mm; lowest in Jan at 5mm<br><br>wet season from May to September; drier months from Oct to March | |
| | **(b)** | Account for the climatic characteristics of Dhaka, Bangladesh as shown in Resource 5. | [6] |
| | | Possible points:<br>**Aw with some monsoon influence (Dhaka is actually Aw climate)**<br>- ITCZ and STH alternating influence on Dhaka → ITCZ low pressure belt → convectional uplift and rainfall in summer months in the middle of the year<br>- Onset of STH when ITCZ shifts south → high pressure belt suppresses formation of clouds and inhibiting rainfall development → drier winter months<br>- Also experiences some influence during the wet summer monsoon and dry winter monsoon (see explanation below)<br><br>**Also accept monsoon circulation explanation**<br>- Moist onshore SW monsoon winds that pass over the Bay of Bengal in the middle of the year driven by shifting ITCZ when sun is overhead in the Northern Hemisphere at the Tropic of Cancer in June → account for wet season + wettest month<br>- Conversely dry offshore winds pass over Dhaka at the end of the year when sun is overhead at Southern Hemisphere.<br><br>**High temp with seasonality** | |

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| | | - **Low latitude** → high temp: Its low latitudinal location means that angle of incidence of the sun is high; Insolation is concentrated over a smaller surface area, leading to its high MAT.<br>- **Shifting overhead sun driven by tilt of earth and orbit around the Sun accounts of temp variation:**<br>o sun overhead at Tropic of Cancer in the middle of the year → longer days → higher temp;<br><br>sun overhead at Tropic of Capricorn towards the end of the year → lower temp + shorter days | |
| | **(c)** | Suggest how the river discharge shown in Resource 6 may influence the development of fluvial landform(s). | [4] |
| | | Variable discharge + erodible banks → braided channel development<br>During high Q phase in months of July to October + data → increase Rh → increased erosivity → increased erosion and transportation → increased sediment load + widening of channel banks<br>Lower Q in months of Dec to June + data → lower Rh → fall in competence and capacity → deposition → accumulation of depositions and intertwining thread of water → braided channel<br>Reserve 1 m for using data | |
| | **(d)** | With reference to Resource 7, describe the track and development of Tropical Cyclone Remal. | [4] |
| | | - First observed in North of Bay of Bengal on 24 May 2024, midnight<br><br>- Evolved quickly into a Cyclonic Storm by 25 May 6 am.<br><br>- Took on a <u>northerly track</u> as it increased in intensity to become a Severe Cyclonic Storm within a day (26 May 2024, 6 am)<br><br>- Made landfall to the west of Khepupara <u>between 26 May, 6 pm and 27 May, 6 am</u> as a Severe Cyclonic Storm<br><br>- Continued northerly track but losing intensity along the way – downgraded to a Deep Depression by 27 May 2024, 6 pm to the <u>west of Dhaka</u>. Dissipated by 28 May 24, 6 am as a Depression <u>north of Dhaka</u> | |
| | **(e)** | Explain how tropical cyclones develop. | [6] |
| | | The warm sea surface of tropical seas / oceans heats up the air above, causing very large parcels of moist air to rise up into the atmosphere. Because of the intense heating over the ocean, the air contains large amounts of water vapour. [1m]<br>As the air rises, water vapour condenses into clouds which releases latent heat of condensation into the surrounding air. [1m]<br>Vast amount of latent heat of condensation cause air to become even warmer than the surrounding atmosphere causing it to rise and fuelling greater | |

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| | | instability and causing more uplift of warm moist surface air above the ocean (convectional uplift). [1m]<br>More winds also converge to these low-pressure areas (convergent uplift), creating a positive feedback loop that causes the storm to intensify and wind speeds to keep increasing. [1m]<br>Coriolis force causes the winds to deflect and rotate as they converge and rise. [1m]<br>Once wind speeds reach 119km/h the storm is considered a tropical cyclone. This cycle continuously repeats itself until the tropical cyclone passes over cool water, land or there is high wind shear to weaken it. [1m]<br>Due to the high pressure present at the upper troposphere, most of the rising air flows outward near the top of the storm. A small proportion of the air sinks towards the centre of the storm and warms adiabatically, creating an area of light winds and clear skies Once the tropical cyclone reaches maturity, a central eye develops. [1m] | |
| | **(f)** | With reference to Resource 8, explain how impacts of fluvial floods could have differed between 1989 to 2009 | [6] |
| | | - Previously extensive agricultural land → floods possibly brought about **positive impacts** → supports agriculture by naturally irrigating fields and depositing nutrient-rich sediments that improves soil fertility.<br><br>- Overtime, increase in urban density and cover → flooding possibly **increased in magnitude due to increased HOF** stemming from loss of vegetation cover (significant decrease: from 50% in cover in 1989 to less than 10%) + replacement with concrete surfaces.<br><br>- **increased economic and social losses**, especially since Dhaka is the Capital city → explain how economic productivity may diminish, increased exposure to flood risk in terms of impacts that result in social and economic losses<br><br>increased urban area also mean that **more flood protection measures in place** → while probability may increase from increase HOF, flood embankments (a form of flood protection measure seen in R8) can limit the impacts in those areas | |

## Section B

Answer **either** question 3 **or** question 4 and answer **either** question 5 **or** question 6.

### Cluster 1: Development, Economy and Environment

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| **3** | 'Thomas Malthus' views on population resource relationship is correct.'<br><br>Evaluate this statement. | [20] |
| | Candidates can make an evaluation on Malthus theory regarding population resource relationship, highlighting how in some LDCs it is still relevant in showing that resources are the main factors limiting population growth. Candidates can then make an assessment to the current world context is malthus theory still rings true particularly when human ingenuity manages to create much more excess food supply and resources. Through a consideration of Boserup's theories, they could use relevant example as a counterpoint. They can also think about how malthus have failed to consider the unevenness of resource distribution as the limits to population growth | |

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| **4** | 'Transnational Corporations (TNCs) does more good than harm to countries where they are located.'<br><br>Evaluate this statement. | [20] |
| | Candidates should first understand that this question encompasses both home and host countries for TNCs. Usage of the right terminology can help reflect strong understanding of question interpretation. Candidate can then evaluate the various social, economic, environmental impacts of TNCs and highlight the condition as to why some impacts are more negative and severe in some countries, while some countries reap more benefits from the existence of TNCs. | |

### Cluster 2: Tropical Environments

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| **5** | Evaluate the view that natural factors are the most important in influencing the movement of materials on slopes. | [20] |

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| *Candidates could approach the question by making a judgement on whether the movement of materials on slopes is influenced by natural factors through a consideration of the relative significance of natural factors vis-à-vis human factors. Candidates could analyse natural factors such as prolonged rainfall and earthquakes, which could trigger a sudden movement of materials. Candidates could also analyse both natural and human factors which may contribute to movement of materials on slopes over time such as weathering and deforestation. Candidates could even analyse human factors such as slope management strategies which could reduce the occurrence of movement of materials on slopes.*<br><br>*Candidates could also approach the question by making a judgment onwhether some factors have a greater influence in some contexts than others through a consideration of two or more case studies. Candidates could evaluate the relative significance of the different factors in each case study and analyse the interplay between the factors.* | | |
| **6** | Evaluate the influence of different factors in the formation of Karst landforms. | [20] |
| | *Candidates could approach the question by making a judgement on the relative significance of factors influencing the formation of Karst landforms such as cockpit karst, tower karst, caves and speleothems. Candidates could analyse the role of natural factors such as climate, geology, vegetation and relative height of water table as well as human factors in the development of Karst landforms*<br><br>*Candidates could also approach the question by making a judgment on whether some factors have a greater influence in some contexts than others through a consideration of two or more case studies. Candidates could evaluate the relative significance of the different factors in each case study and analyse the interplay between the factors.* | |

**[End of Paper]**